

Supervisor (Supervised) Machine Learning

UNIT-I: Introduction to Supervised Learning

1. Define Supervised Learning. How is it different from Unsupervised Learning?
2. Explain the steps in the Supervised Learning workflow.
3. Discuss the role of training and testing data in supervised learning.
4. What are regression and classification problems? Give examples.

UNIT-II: Linear and Logistic Regression

1. Derive the formula for simple linear regression.
2. Explain cost function and gradient descent with respect to linear regression.
3. How does logistic regression work for classification?
4. Explain sigmoid function and its significance.
5. Compare Linear Regression vs Logistic Regression.

UNIT-III: Decision Trees and Random Forests

1. Describe how a decision tree is constructed.
2. Explain Gini Index and Information Gain with examples.
3. What is pruning in decision trees? Why is it needed?
4. Describe the working of Random Forest algorithm.
5. Compare Decision Tree and Random Forest with advantages and disadvantages.

UNIT-IV: Support Vector Machines (SVM)

1. What is the objective of SVM? Explain the concept of hyperplane.
2. Define margin, support vectors and explain their role in SVM.
3. Discuss kernel trick with examples (Polynomial, RBF).
4. Compare linear vs non-linear SVM.
5. Advantages and limitations of SVM.

UNIT-V: K-Nearest Neighbors (KNN) and Naïve Bayes

1. Explain the working of KNN algorithm with an example.
2. What are the advantages and limitations of KNN?
3. Discuss the distance metrics used in KNN (Euclidean, Manhattan).
4. Explain Naïve Bayes classifier with an example.
5. What are the assumptions of Naïve Bayes? When does it fail?